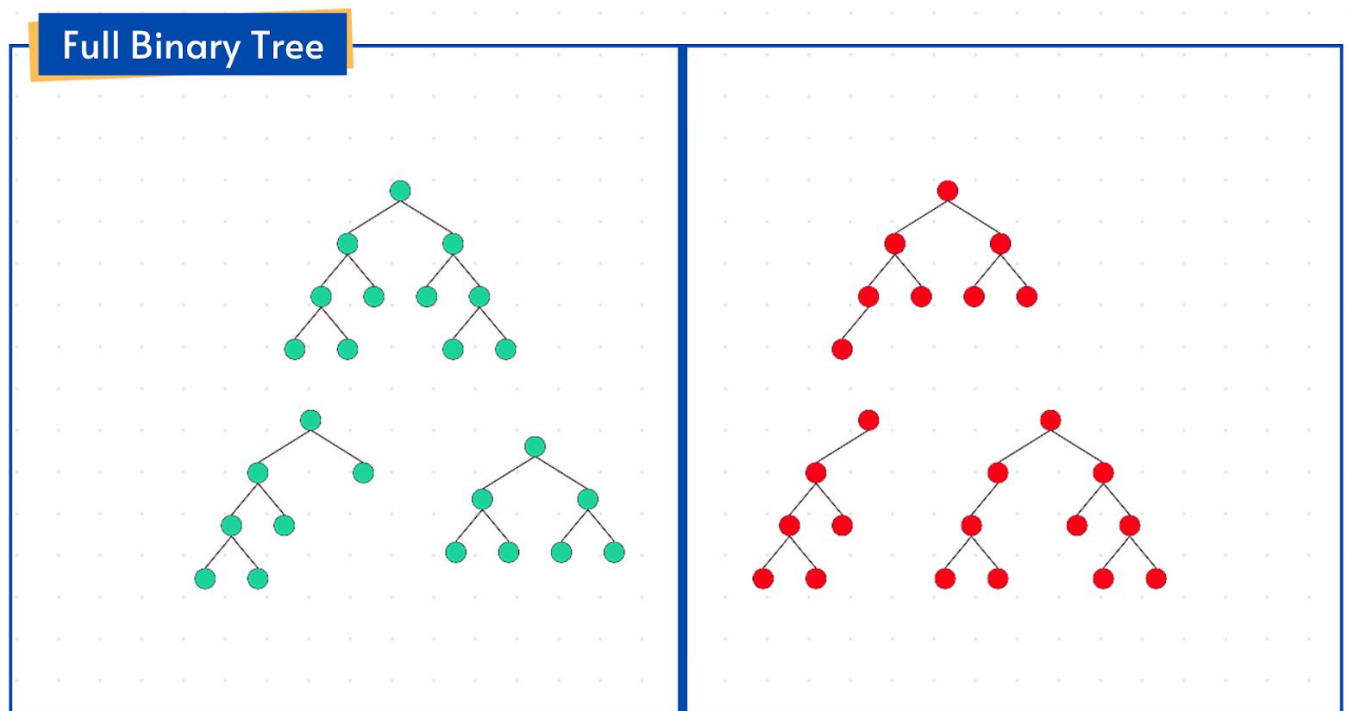


Types of Binary Tree

A binary tree is a tree data structure composed of nodes, each of which has at most, two children, referred to as left and right nodes. Let us understand 5 types of Binary Tree through definitions and with their valid-invalid structures.

1. Full Binary Tree

Full Binary Tree is a Binary Tree in which every node has 0 or 2 children.



Side Note: For Full Binary Tree, the following equation is always true.

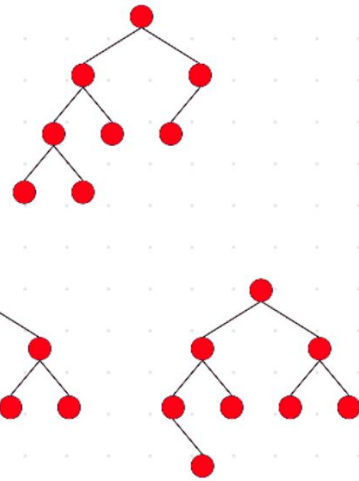
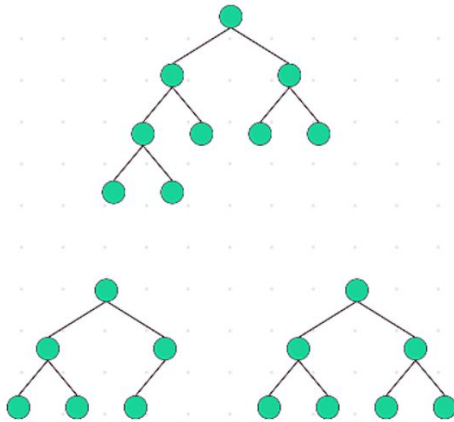
$$\text{Number of Leaf nodes} = \text{Number of Internal nodes} + 1$$

2. Complete Binary Tree

It is a binary tree in which every level, except possibly the last, is completely filled, and all nodes are as far left as possible.

Side Note: Binary Heap is an important use case of Complete Binary tree.

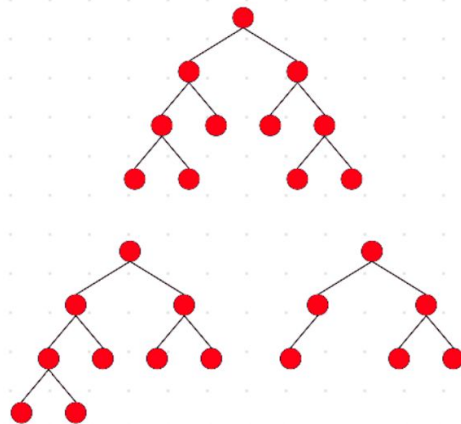
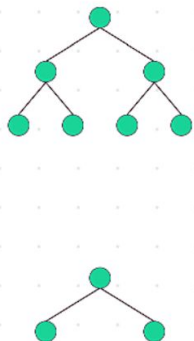
Complete Binary Tree



3. Perfect Binary Tree

Perfect Binary Tree is a Binary Tree in which all internal nodes have 2 children and all the leaf nodes are at the same depth or same level.

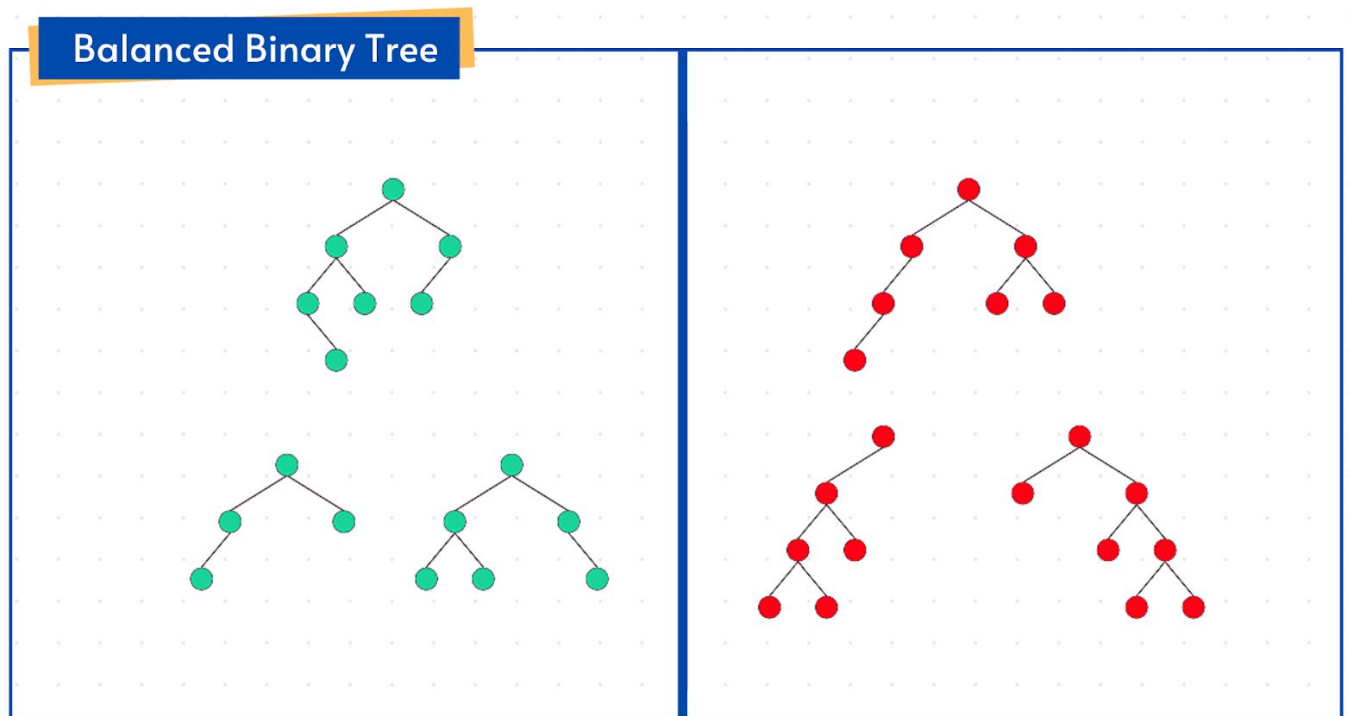
Perfect Binary Tree



Side Note: Total number of nodes in a Perfect Binary Tree with height H is $2^H - 1$.

4. Balanced Binary Tree

Balanced Binary Tree is a Binary tree in which the height of the left and the right subtrees of every node may differ by at most 1.

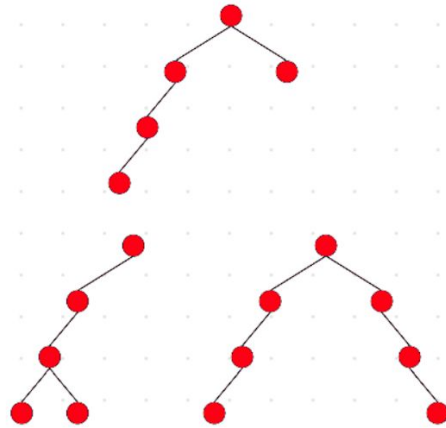
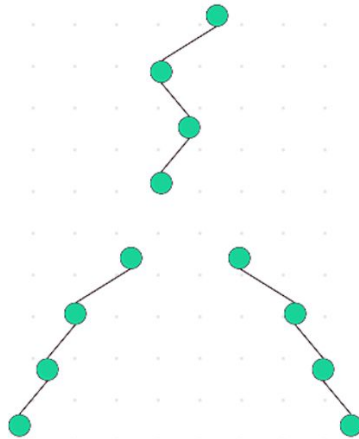


Side Note: AVL Tree and Red-Black Tree are well-known data structures to generate/maintain Balanced Binary Search Tree. Search, insert and delete operations cost $O(\log n)$ time in that. We will discuss about both of them later in the course.

5. Degenerate Binary Tree

It is a tree where each parent node has only one child node. It behaves like a linked list.

Degenerate Binary Tree



Side Note: Height of a Degenerate Binary Tree is equal to Total number of nodes in that tree.